MIL-D-409C (TP)
9 January 1970
SUPERSEDING
MIL-D-409B
17 June 1965

MILITARY SPECIFICATION

DETONATORS, PERCUSSION, DELAY, 15 SECOND, M1A2 AND 8 SECOND, M2A1 LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of delay detonator designated as Detonator, Percussion, Delay, 15 Second, M1A2 and 8 Second, M2A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-C-13739	-	Composition Delay

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defectives
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8799903 - Detonator, Percussion Delay 15 Sec. M1A2 and 8 Sec. M2A1
8825212 - Carton, Packing, Ammunition, for Detonator Delay M1A1 or M2A1.
8825213 - Box, Packing, Ammunition, for Detonator, Delay M1A1 or M2A1

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET-8799903 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonator.-The detonator shall comply with all requirements specified on drawing (dwg.) 8799903, and with all requirements specified in applicable specifications.

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3.3 Moisture content of delay composition.-The moisture content of the delay composition at the loading station at the time of loading the detonator shall not exceed the amount specified in MIL-C-13739, when tested as specified in 4.4.1.

3.4 Environmental Tests of Delay Housing and Primer Holder Assembly.-Neither primer nor primer holder assembly shall blow out in functioning after being subjected to the environmental tests specified in MIL-STD-331. The test shall be performed as specified in 4.4.5.

3.5 Functioning.-The delay detonator shall function on the first attempt and shall comply with the following requirements.

3.5.1 At ambient temperature (70 plus or minus 10 degrees Fahrenheit (°F)).-Prior to functioning, the delay detonator shall be submerged in water to a minimum depth of three (3) feet for not less than six (6) hours. Within five minutes after removal from the water, the detonator shall be subjected to the functioning test as specified in 4.4.7.

3.5.1.1 M1A2.-The M1A2 detonator shall function in not less than 13 seconds nor more than 18 seconds.

3.5.1.2 M2A1.-The M2A1 detonator shall function in not less than 7.5 seconds nor more than 10 seconds.

3.5.2 At -65°F

3.5.2.1 M1A2.-The M1A2 detonator shall function when tested as specified in 4.4.7.

3.5.2.2 M2A1.-The M2A1 detonator shall function when tested as specified in 4.4.7.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

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3.7 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a sample of 100 delay detonators, 100 delay housing and primer holder assemblies, 50 firing pin assemblies and 50 blasting caps to the Government for testing in accordance with the provisions of 4.2.2. All parts and materials shall be the same as will

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be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.-Assemblies, subassemblies, and components will be subjected by the Government to any or all of the examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawings. Assemblies will be inspected 100% for presence of a delay column and then the following tests shall be performed.

a. Push out force of primer holder assembly.-50 delay housing and primer holder assemblies shall be tested as specified in 4.4.6.

b. Environmental tests of delay housing and primer holder assembly.-50 assemblies shall be tested as specified in 4.4.5.

c. Functioning at ambient temperature.-50 delay detonators shall be tested as specified in 4.4.7.

d. Functioning at -65°F.-50 delay detonators shall be tested as specified in 4.4.7.

4.2.3 Rejection.-If any assembly fails to comply with the applicable requirements, the first article sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as

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specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of detonators shall contain:

a. Metal parts from not more than one lot interfix number from one manufacturer.
b. Delay composition from not more than two consecutive batches from one manufacturer.
c. Blasting caps from not more than one lot interfix number from one manufacturer.
d. Primers from not more than one lot interfix number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Delay housing prior to loading (see dwg. 8799915 covering a detail of dwg. 8799903)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Flash hole obstructed	Visual	01001

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Major: None defined

Minor:		AQL 0.65 Percent	
201.	Evidence of poor workmanship (See 3.6)	Visual	01002

4.3.2.2 Primer holder assembly (see dwg. 8799916 covering a detail of dwg. 8799903)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Primer above flush	Gage	11001
102.	Crimp missing or incomplete	Visual	11003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	11002

4.3.2.3 Delay housing and primer holder assembly (see dwg. 8799911 covering a detail of dwg. 8799903)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Different delay marking in same lot	Visual	02001
Major:	AQL 0.65 Percent		
101.	Primer missing	Visual	02002
102.	Crimp or stake missing or incomplete	Visual	02003
103.	Sealer missing or inadequate	Visual	02004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02006

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4.3.2.4 Detonator, delay assembly (see dwg. 8799908 covering a detail of dwg. 8799903)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Blasting cap cracked or split	Visual	03001
Major:	AQL 0.65 Percent		
101.	Blasting cap crimp improper or loose	Visual/Manual	03002
102.	Sealer missing or inadequate	Visual	03003
103.	Firing pin housing assembly stake inadequate	Visual	03004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship, (See 3.6)	Visual	03005

4.3.2.5 Assembly (See dwg. 8799903)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Incorrect pull ring in relation to marking	Visual	04001
2.	Cotter pin(s) not spread or missing	Visual	04002
Major:	AQL 0.65 Percent		
101.	Protector cap missing	Visual	04003
102.	Marking missing, misleading or unidentifiable	Visual	04004
103.	Color or marking incorrect	Visual	04005
104.	Safety pin pull cord missing	Visual	04006
Minor:	AQL 1.00 Percent		
201.	Total length	Gage	04007
202.	Evidence of poor workmanship (See 3.6)	Visual	04008

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4.3.2.6 Carton, Inner (See dwg. 8825212)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Carton damaged	Visual	05001
102.	Number of detonators incorrect	Visual	05002
Minor:	AQL 1.00 Percent		
201.	Marking misleading or unidentifiable	Visual	05003
202.	Any packing material missing	Visual	05004

4.3.2.7 Carton, Outer (See dwg. 8825212)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Number of inner cartons incorrect	Visual	06001
102.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal the carton completely	Visual	06002
Minor:	AQL 1.50 Percent		
201.	Carton torn, cut or punctured	Visual	06003
202.	Marking misleading or unidentifiable	Visual	06004
203.	Contents move when carton is shaken	Manual	06005

4.3.2.8 Sealed barrier bag (see dwg. 8825212)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 Percent		
101.	Seal missing, incomplete or incorrectly located	Visual	07001
102.	Bag punctured, torn or cut	Visual	07002

Minor: None defined

4.3.2.9 Box, packing, ammunition (see dwg. 8825213)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Board broken or split	Visual	08001
102.	Strapping missing, broken or loose	Visual	08002
Minor:	AQL 1.50 Percent		
201.	Marking misleading or unidentifiable	Visual	08003
202.	Strapping improperly assembled	Visual	08004
203.	Contents move when box is tilted	Manual	08005

4.3.3 Testing

4.3.3.1 Moisture content of delay composition (see 3.3), Major Defect, Code No. 09001.-Sampling plans and procedures for this test shall be as specified in MIL-C-13739. This test shall be performed as specified in 4.4.1.

4.3.3.2 Column height of delay housing assembly (see dwg. 8799913 or dwg. 8799914 as applicable), Major Defect, Code No. 09002.-This inspection shall be conducted 100 percent. Any assembly failing to meet the requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using test equipment specified in 4.3.4.

4.3.3.3 Surface of delay charge from bottom of delay cavity (see dwg. 8799913 or dwg. 8799914 as applicable), Major Defect, Code No. 09003.-This inspection shall be conducted 100 percent. Any delay housing assembly failing to meet the requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3 using test equipment specified in 4.3.4.

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4.3.3.4 Presence of delay column (see dwg. 8799913 or dwg. 8799914 as applicable), Critical Defect, Code No. 09004.-This examination shall be conducted 100 percent. Any assembly which contains one or more of the following defects shall be classed defective and removed from the lot:

a. Delay column missing.
b. Delay column not uniformly consolidated.
c. Voids in delay column.
d. Height of delay column less than 1.46 inch when measured from the top of the anvil punch mark or breakout (detonator end).
e. Anvil punch penetration at the bottom of the delay column, including any breakout, that extends more than .100 inch up into the column (detonator end).

The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.5 Environmental tests of delay housing and primer holder assembly (see 3.4), Major Defect, Code No. 09005.-A sample of 50 assemblies shall be selected from each lot. If one or more assembly fails to comply with the drawing requirement specified, the lot shall be rejected. In the event a functioning time of less than 13 seconds for the M1A2 detonator or 7.5 seconds for the M2A1 detonator, as applicable, is observed the lot shall be rejected. The test shall be performed as specified in 4.4.5 using test equipment as specified in 4.3.4.

4.3.3.6 Push out force of primer holder assembly (see dwg. 8799911) Major Defect, Code No. 09006.-A sample of 100 delay housing and primer holder assemblies shall be selected from each lot for this test. The lot shall be rejected if any primer holder assembly fails to meet the requirement. The test shall be performed as specified in 4.4.6 using test equipment as specified in 4.3.4.

4.3.3.7 Functioning

4.3.3.7.1 Ambient temperature (see 3.5.1), Major Defect, Code No. 09007.

4.3.3.7.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with acceptance criteria specified, the sampling plans for this test shall be in

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accordance with MIL-STD-414, Section B, Table B-3 using Code Letter M and AQL 1.0 percent. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4. If two (2) detonators fail to function on the first attempt the lot shall be rejected. If one (1) detonator fails to fire on the first attempt, the following two (2) actions shall be taken:

1. For the sampling plan calculations, another delay detonator shall be selected from the lot and tested as required in 4.4.7.
2. Twenty-five (25) additional sample delay detonators shall be selected from the lot and functioned as specified in 4.4.7 omitting the delay time recording. If any detonator fails to fire on the first attempt, the lot shall be rejected.

4.3.3.7.1.2 After three consecutive lots have met the criteria of 4.3.3.7.1.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3 using Code Letter H and AQL 1.0 percent. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4. If two (2) detonators fail to function on the first attempt, the lot shall be rejected. If one (1) detonator fails to fire on the first attempt, the following two (2) actions shall be taken:

1. For the sampling plan calculations, another delay detonator shall be selected from the lot and tested as required in 4.4.7.
2. Twenty-five (25) additional sample delay detonators shall be selected from the lot and functioned as specified in 4.4.6 omitting the delay time recording. If any detonator fails to fire on the first attempt, the lot shall be rejected.

4.3.3.7.2 -65°F (See 3.5.2) Major Defect, Code No. 09008.-A single sampling plan shall be used for this test in accordance with MIL-STD-105, using Code Letter J and an AQL of 0.65 percent. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4. The delay times shall be recorded for informational purposes. If any delay detonator fails to function on the first attempt, it shall be classed defective.

4.3.3.8 Check test for deterioration of primers.-If the total time between original acceptance of any primer lot and the assembly of that lot into the delay detonators exceeds two years, or if the primers have been subjected to adverse conditions,

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however brief, at any time since previous tests, the primer lot shall be subjected to, and must satisfactorily pass, the check test specified in the applicable primer loading specification. The test shall be performed by the contractor of the facilitiy assembling the primers into the delay detonators, on samples selected by the Government inspector, immediately before the primers are assembled into the delay detonators. The test shall be conducted at Government expense without cost to the contractor (see 6.4) who loaded the primers or the contractor assembling the primers into the delay detonators, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET-8799903 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production, and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content of delay composition.-The moisture content of the delay composition shall be determined as specified in MIL-C-13739.

4.4.2 Column height of delay housing assembly.-Prior to assembly of the delay housing assembly with the primer holder, the height of the delay column shall be determined by using a gage as specified in 4.3.4.

4.4.3 Surface of delay charge from bottom of delay cavity.-Prior to assembly of the delay housing assembly with the detonator, the surface of the delay charge from the bottom of the delay cavity shall be determined using a gage specified in 4.3.4.

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4.4.4 Presence of delay columns.-The final assembly shall be subjected to radiographic inspection on film in accordance with MIL-STD-453. If a defective assembly is found, the cause shall be investigated and corrected. The test method, exposures, records of handling and identification of material undergoing test shall be examined by the Government inspector at regular but unscheduled intervals to determine that the test and observation for defects is being carried out in accordance with these provisions.

4.4.5 Environmental tests of delay housing and primer holder assembly.-The delay housing and primer holder assembly shall be subjected to the following environmental tests as specified in MIL-STD-331 in the sequence indicated.

Test	Test No.
1. Shock Test	113
2. Jolt Test	101
3. Jumble Test	102

Exceptions to MIL-STD-331 for the applicable environmental tests are as follows:

Shock test.-The low and high temperatures shall be minus 30°F and 140°F respectively, and the length of each exposure to each temperature shall be 2 hours. The test shall continue for 2 cycles, each cycle consisting of 1 high temperature and 1 low temperature exposure.

Jumble test.-A maximum of seven assemblies may be tested in one jumble box providing adequate protection is provided to prevent primer firing or damage to the samples.

Upon completion of all tests, the delay housing and primer holder assembly shall be assembled with a firing pin assembly and a blasting cap and functioned in accordance with 4.4.7.

4.4.6 Push out force of primer holder assembly.-The sample delay housing and primer holder assemblies shall be sectioned at a desired length to accommodate a punch. The sectioned piece shall be placed in a holding fixture and the punch inserted through the bottom opening to make contact with the primer holder assembly. A force shall be applied until the primer holder assembly shows evidence of moving. The force required to move the primer holder assembly shall be recorded to determine compliance with the applicable requirement.

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4.4.7 Functioning.-The delay detonator shall be mounted in the applicable test fixture specified in 4.3.4 and the timer set at zero. The safety pin shall be removed and the detonator fired by releasing the pull ring. The delay time shall be recorded to the nearest 0.01 second. Observation shall be made to determine compliance with the applicable requirement. The delay detonators to be functioned at -65°F shall be conditioned at the required temperature for 8 hours ± 1 hours and shall be tested within 10 minutes after leaving the conditioning chamber.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The detonator shall be packaged in accordance with dwg. 8825212.

5.2 Packing

5.2.1 Level A.-The packing carton shall be packed in accordance with dwg. 8825213.

5.3 Marking.-Marking shall be in accordance with dwg. 8825212 and 8825213.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all the

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information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Check test expense.-The contracting officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-060

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